

A.1 One-Constant Theory-Guided Spectral Shrinkage

Using past-only raw managed eigenvalues $\widehat{\mu}_{j,y}$ of $g_y g_y^\top / T_y$, consider

$$\widehat{C}_y(\lambda) = \sum_j \frac{\widehat{\mu}_{j,y}}{\widehat{\mu}_{j,y} + \lambda}, \quad \widehat{\lambda}_y = \arg \min_{\lambda \in [\lambda_{\min}, \lambda_{\max}]} \left\{ \lambda + \frac{\rho_0}{T_y} \widehat{C}_y(\lambda) \right\}. \quad (1)$$

The original 120-point grid fixes the bounds. Strict convexity permits Brent root finding, with boundary selection by derivative signs:

$$\rho_0 = \frac{T_0}{\sum_j \widehat{\mu}_{j,0} / (\widehat{\mu}_{j,0} + \lambda_0)^2}, \quad 1 = \frac{\rho_0}{T_y} \sum_j \frac{\widehat{\mu}_{j,y}}{(\widehat{\mu}_{j,y} + \widehat{\lambda}_y)^2}. \quad (2)$$

Calibration uses only the 1978 decision (training 1963–1972, validation 1973–1977, refit $T_0 = 180$): $\lambda_0 = 2.515 \times 10^{-7}$, $\rho_0 = 4.344 \times 10^{-6}$. The ratio is frozen thereafter; first-year agreement holds by construction.

With identical 10,000 Gaussian features, bandwidth, annual splits, ridge solver and scale, the 552 common returns for decisions 1979–2024 give CV/spectral gross Sharpe 3.277/3.330. The difference +0.053 has paired 95% interval [-0.285,+0.377] (12-month circular blocks, 5,000 draws, conditional on fitted paths). Reconstructed 25-bp target-turnover plus 30-bp/year short-fee Sharpes are 2.597/2.686. There are 0 boundary solutions; grid-rule Sharpe is 3.332. We recommend excluding this method: financial improvement is not convincing.

The fixed-penalty upper bound motivates this proposal without guaranteeing oracle optimality. Data-dependent selection requires uniform-in-penalty concentration; historical calibration need not be structural. No polynomial Gaussian tail is assumed. Finite sample rank and nonstationarity remain limitations.